Chapter 1. Introduction

Experiment, Design, and Statistics

1 Experiment, Design, and Statistics

- An experiment aims to **draw inferences** or **make a decision** about a hypothesis, which needs **data**.
- *Example (umbrella).* Barometer, sky, forecast, radio: data assimilated into a decision.
- **Statistics**: making decisions in the presence of variability (uncertainty); **how the data are collected** is therefore crucial.
- **Experimental designs** reduce experimental error; **randomization** averages out uncontrolled variables.

Definition 1.1: Basic principles of experimental design

(i) **Randomization.** Allocation of material and run order are random. This usually validates the assumption of independent errors and "averages out" extraneous (lurking) factors.

(ii) **Replication.** An independent repeat run of each factor combination. It (a) allows estimation of the experimental error σ^2, the yardstick for real differences, and (b) sharpens means: n replicates give variance σ^2/n.

(iii) **Blocking.** Improves precision of comparisons by removing variability from nuisance factors; a block is a set of relatively homogeneous conditions.

Remark 1.2: Why replication shrinks variance

If $y_1, \dots, y_n$ are independent with common variance σ^2, then

$$\begin{aligned}\operatorname{Var}(\bar{y}) &= \operatorname{Var}\Big(\frac{1}{n}\sum_{j=1}^{n} y_j\Big) = \frac{1}{n^2}\operatorname{Var}\Big(\sum_{j=1}^{n} y_j\Big) = \frac{1}{n^2}\sum_{j=1}^{n}\operatorname{Var}(y_j) \quad \text{(independence)} \\ &= \frac{1}{n^2}\cdot n\sigma^2 = \frac{\sigma^2}{n}.\end{aligned}$$

Used in every later ANOVA derivation, e.g. $\operatorname{Var}(\bar{\varepsilon}_{i\cdot}) = \sigma^2/n$.

2 Role of Statistics in Experimentation

- **Project planning phase**
  1. What is to be measured? (choice of the response variable)
  2. How large is the likely variation?
  3. What are the influential factors?
- **Experimental design phase**
  1. Control some known sources of variation (blocking).
  2. Allow estimation of the size of the uncontrolled variation (replication).

3. Permit an investigation of suitable models.

- **Statistical analysis phase**

1. Make inferences on design factors.
2. Guide subsequent design.
3. Suggest more appropriate models.

3 The General Workflow Used in This Course

Derivation

Every design is analyzed in three steps (two-sample model $y_{ij} = \mu_i + \varepsilon_{ij}$, $\varepsilon_{ij} \sim N(0, \sigma^2)$, i.e. $y_{ij} \sim N(\mu_i, \sigma^2)$, $H_0 : \mu_1 = \mu_2$):

① **Estimation**: fit the model, $y_{ij} = \hat{\mu}_i + e_{ij}$ (fitted value + residual);

② **Inference**: test hypotheses / build confidence intervals on the parameters;

③ **Residual analysis**: check the model assumptions (normality, constant variance, independence) using the residuals e_{ij}.

Design	Model	Chapter
Two-sample / paired comparison	$y_{ij} = \mu_i + \varepsilon_{ij}$, $y_{ij} = \mu_i + \beta_j + \varepsilon_{ij}$	2
Completely randomized design (CRD)	$y_{ij} = \mu + \tau_i + \varepsilon_{ij}$	3
RCBD / BIBD	$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$	4
Latin / Graeco-Latin square	$y_{ijk} = \mu + \alpha_i + \tau_j + \beta_k + \varepsilon_{ijk}$	4
Factorial design	$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$	5
2^k factorial, blocking & confounding	effects via contrasts	6, 7
Regression approach	$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$	10
Random / mixed effects	$\tau_i \sim N(0, \sigma_\tau^2)$	13
Nested and split-plot	$y_{ijk} = \mu + \tau_i + \beta_{j(i)} + \varepsilon_{(ij)k}, \ldots$	14
Analysis of covariance	$y_{ij} = \mu + \tau_i + \beta(x_{ij} - \bar{x}_{\cdot\cdot}) + \varepsilon_{ij}$	15

4 Selected exercises

Motivation. Short calculations behind randomization and replication.

Exercise 1.1 (Randomization makes the comparison unbiased)

$N = n + m$ experimental units have fixed unit effects $u_1, \ldots, u_N$ (unknown). A completely randomized design assigns a simple random sample A of n units to treatment 1 and the other m units (B) to treatment 2, and unit k responds $y_k = u_k + \tau_1$ or $y_k = u_k + \tau_2$ (additivity). With $\bar{u} = \frac{1}{N}\sum_k u_k$ and $S_u^2 = \frac{1}{N-1}\sum_k (u_k - \bar{u})^2$, show that, over the randomization,

$$\mathrm{E}(\bar{y}_A - \bar{y}_B) = \tau_1 - \tau_2, \qquad \mathrm{Var}(\bar{y}_A - \bar{y}_B) = S_u^2\Big(\frac{1}{n} + \frac{1}{m}\Big).$$

Solution [Supplement].

Step 1 (reduction). $\bar{y}_A - \bar{y}_B = \tau_1 - \tau_2 + \bar{u}_A - \bar{u}_B$, and since $n\bar{u}_A + m\bar{u}_B = N\bar{u}$,

$$\bar{u}_A - \bar{u}_B = \bar{u}_A - \frac{N\bar{u} - n\bar{u}_A}{m} = \frac{N}{m}(\bar{u}_A - \bar{u}).$$

Step 2 (indicators). $I_k = \mathbf{1}\{k \in A\}$: $\mathrm{E}I_k = n/N$ and, for $k \neq l$, $\mathrm{E}(I_k I_l) = \frac{n(n-1)}{N(N-1)}$, so

$$\mathrm{Var}I_k = \frac{n}{N}\Big(1 - \frac{n}{N}\Big), \qquad \mathrm{Cov}(I_k, I_l) = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = -\frac{n(N-n)}{N^2(N-1)}.$$

Step 3 (mean). $\bar{u}_A = \frac{1}{n}\sum_k I_k u_k$ gives $\mathrm{E}\bar{u}_A = \frac{1}{n}\sum_k \frac{n}{N} u_k = \bar{u}$, so $\mathrm{E}(\bar{y}_A - \bar{y}_B) = \tau_1 - \tau_2$.

Step 4 (variance). Write $v_k = u_k - \bar{u}$, so $\sum_k v_k = 0$ and $\bar{u}_A - \bar{u} = \frac{1}{n}\sum_k I_k v_k$. Using $\sum_{k\neq l} v_k v_l = (\sum_k v_k)^2 - \sum_k v_k^2 = -\sum_k v_k^2$,

$$\begin{aligned}\mathrm{Var}\Big(\sum_k I_k v_k\Big) &= \frac{n(N-n)}{N^2}\sum_k v_k^2 + \frac{n(N-n)}{N^2(N-1)}\sum_k v_k^2 \\ &= \frac{n(N-n)}{N(N-1)}\sum_k v_k^2 = \frac{n(N-n)}{N} S_u^2.\end{aligned}$$

Hence $\mathrm{Var}(\bar{u}_A) = \frac{N-n}{nN} S_u^2 = \frac{m}{nN} S_u^2$ and, by Step 1,

$$\mathrm{Var}(\bar{y}_A - \bar{y}_B) = \frac{N^2}{m^2} \cdot \frac{m}{nN} S_u^2 = \frac{N}{nm} S_u^2 = S_u^2\Big(\frac{1}{n} + \frac{1}{m}\Big).$$

No distributional assumption was used: randomization itself creates the "error"; S_u^2 plays the role of σ^2 in Ch. 2. ■

Exercise 1.2 (A hidden time trend)

Two treatments are each run n times in $N = 2n$ consecutive runs, and the response drifts linearly with run order k: $y_k = \gamma k + \tau_{(k)} + \varepsilon_k$ (γ unknown, $\tau_{(k)}$ the treatment of run k, ε_k iid with mean 0).
(a) If all treatment-1 runs come first, find the bias of $\bar{y}_1 - \bar{y}_2$ for $\tau_1 - \tau_2$.
(b) If the run order is randomized, show the estimator is unbiased and find the variance contributed by the trend.

Solution [Supplement].

(a) Average run indices are $(n+1)/2$ and $n + (n+1)/2$, so $\mathrm{E}(\bar{y}_1 - \bar{y}_2) = \tau_1 - \tau_2 - \gamma n$: a bias $-\gamma n$ confounded with the treatment difference.

(b) Exercise 1.1 with $u_k = \gamma k$ (errors average out independently of the assignment): unbiased, and with $\sum_{k=1}^{N}(k - \frac{N+1}{2})^2 = \frac{N(N^2-1)}{12}$,

$$S_u^2 = \gamma^2 \frac{N(N+1)}{12}, \qquad \mathrm{Var}_{\mathrm{trend}} = S_u^2 \cdot \frac{2}{n} = \gamma^2 \frac{2n(2n+1)}{12} \cdot \frac{2}{n} = \gamma^2 \frac{2n+1}{3}.$$

Bias becomes random error; blocking on time would remove it. ■

Exercise 1.3 (Replicates versus repeated measurements)

Suppose $y_1, \dots, y_n$ each have variance σ^2, but every pair has correlation $\rho \geq 0$ (e.g. n measurements on one experimental unit instead of n units).
(a) Show $\mathrm{Var}(\bar{y}) = \frac{\sigma^2}{n}\{1 + (n-1)\rho\}$ and that it tends to $\rho\sigma^2$, not to 0, as $n \to \infty$.
(b) Show that $\mathrm{E}(S^2) = (1-\rho)\sigma^2$, so the sample variance underestimates the variability that matters for comparing units.

Solution [Supplement].

(a) $\mathrm{Var}(\sum_j y_j) = \sum_j \mathrm{Var}(y_j) + \sum_{j \neq l} \mathrm{Cov}(y_j, y_l) = n\sigma^2 + n(n-1)\rho\sigma^2$; divide by n^2. The limit is $\rho\sigma^2 > 0$ when $\rho > 0$.

(b) $\mathrm{E}\sum_j (y_j - \bar{y})^2 = \sum_j \mathrm{E}(y_j - \mu)^2 - n\mathrm{E}(\bar{y} - \mu)^2 = n\sigma^2 - \sigma^2\{1 + (n-1)\rho\} = (n-1)(1-\rho)\sigma^2$.

Repeated measurements shrink the apparent error and stop reducing $\mathrm{Var}(\bar{y})$; only independent replicates give σ^2/n and an honest σ^2. ■